

Submission result

A four-stage DVC pipeline trains a fully connected ANN on Fashion-MNIST. The final tagged experiment exceeds the 85% target and the complete unsquashed history demonstrates the required Git operations.

V1 TEST ACCURACY

87.41%

128 units

V2 TEST ACCURACY

87.72%

256 units

PIPELINE STATE

Clean

dvc status

Git history and branch workflow

```
$ git log --oneline --graph --all
```

```
* f1a4910 (HEAD -> main, origin/main) merge: add final documentation
|\
| * 941ec73 (origin/dev, dev) docs: add reproducibility guide and experiment results
|/
* ee2871e merge: resolve simulated code and DVC data conflicts
|\
| * 9a7b3df (origin/teammate-sim, teammate-sim) collab: simulate teammate normalization and data version
| * | 47614a4 collab: add independent main normalization and data version
|/
* a279595 merge: integrate v2 experiment
|\
| * 22e7d39 (tag: v2) experiment: train v2 with 256 dense units
|/
* a1373e4 merge: integrate reproducible v1 pipeline
|\
| * 349dc79 pipeline: migrate v1 artifacts to four-stage DVC workflow
| * b7e8efc (tag: v1) data: track v1 datasets model and metrics with DVC
| * b9b08d6 chore: isolate ML and DVC environments
```

Required command demonstrations

Part A command/output record

```
$ git log --stat -3           # files and line counts in recent commits
$ git log -p -1              # exact patch introduced by the latest commit
$ git log main..dev          # commits reachable from dev but not main
$ git diff                   # unstaged preprocess.py range validation
$ git diff --staged          # same change after staging
$ git diff main..dev         # endpoint comparison
$ git diff main...dev        # changes since the merge base
$ git stash push -m 'WIP preprocess range validation'
$ git stash list
stash@{0}: On dev: WIP preprocess range validation
$ git stash pop              # restored the exact mid-edit change
$ git reset --hard HEAD~1    # working tree returned clean
$ git reset --soft HEAD~1    # reset_demo.txt remained staged
$ git mv PROJECT_NOTES.md docs/PROJECT_NOTES.md
$ git rm obsolete_scratch.txt
```

Rebase: a README hotfix was committed on a short-lived branch, fast-forwarded into main, and dev was rebased onto it. The graph changed from parallel lines to a clean feature sequence above the hotfix.

Diff semantics: two-dot compares the two branch tips. Three-dot compares the right tip with the best common ancestor, isolating work introduced on dev. In the captured linear case, both outputs matched because main was the merge base.

Modular implementation

Stage	Responsibility	Versioned output
prepare	Load 60k/10k Fashion-MNIST arrays	data/raw
preprocess	Normalize and stratified 90/10 split	data/processed
train	Flatten-Dense-Dropout-Softmax ANN	model.h5 + history.csv
evaluate	Loss, accuracy, predictions and matrix	metrics.json + PNG

Final params.yaml

```
preprocess:
  test_size: 0.1
  seed: 42

train:
  dense_units: 256
  dropout_rate: 0.25
  learning_rate: 0.001
  epochs: 8
  batch_size: 128
  seed: 42
```

Final dvc.yaml 1 of 2

```
stages:
  prepare:
    cmd: python src/prepare.py
    deps:
      - src/prepare.py
    outs:
      - data/raw

  preprocess:
    cmd: python src/preprocess.py
    deps:
      - src/preprocess.py
      - data/raw
    params:
      - preprocess.test_size
      - preprocess.seed
    outs:
      - data/processed

  train:
    cmd: python src/train.py
    deps:
      - src/train.py
```

Final dvc.yaml 2 of 2

```
- data/processed
  params:
    - train.dense_units
    - train.dropout_rate
    - train.learning_rate
    - train.epochs
    - train.batch_size
    - train.seed
  outs:
    - models/model.h5
    - models/history.csv

evaluate:
  cmd: python src/evaluate.py
  deps:
    - src/evaluate.py
    - data/processed/test.npz
    - models/model.h5
  outs:
    - models/confusion_matrix.png
  metrics:
    - metrics.json
    cache: false
```

DVC tracking and Google Drive remote

Part C configuration and pointer evidence

```
$ dvc init
Initialized DVC repository.
$ dvc remote add -d gdrive_storage gdrive://1iy0k52bAA4H6i9_KrBaXYRkjCIn0uMYg
Setting 'gdrive_storage' as a default remote.
$ dvc add data/raw data/processed models
Created data/raw.dvc, data/processed.dvc and models.dvc
$ dvc push --all-branches --all-tags
11 files pushed
$ dvc status -c
Cache and remote 'gdrive_storage' are in sync.

v1 data/raw.dvc
outs:
- md5: 172b17a74a544f1f7617f8a6b2153055.dir
```

Pipeline migration: standalone dvc add created the rubric's .dvc pointers at v1. The same paths were then migrated to dvc.yaml outputs because DVC correctly rejects tracking one path simultaneously as both a standalone pointer and a stage output. Modern pipeline artifact hashes therefore live in dvc.lock.

Final project structure

```
Fashion-ANN-Pipeline/
|- src/{prepare,preprocess,train,evaluate}.py
|- data/{raw,processed}/          [DVC cached]
|- models/                        [DVC cached]
|- params.yaml                    [single source of hyperparameters]
|- dvc.yaml / dvc.lock            [pipeline and exact artifact hashes]
```

D3 first complete reproduction

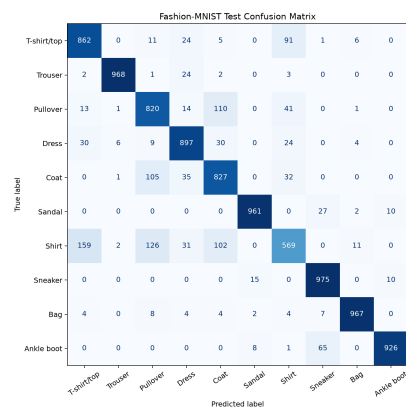
Actual condensed v1 console output

```
$ dvc repro
Running stage 'prepare': > python src/prepare.py
Saved 60,000 training samples and 10,000 test samples
Running stage 'preprocess': > python src/preprocess.py
Saved splits: train=54,000, validation=6,000, test=10,000
Running stage 'train': > python src/train.py
Epoch 8/8 - accuracy: 0.8832 - val_accuracy: 0.8912
Running stage 'evaluate': > python src/evaluate.py
{"test_loss": 0.348672, "test_accuracy": 0.8741, "test_samples": 10000}
Generating and updating dvc.lock
```

D4 parameter change and selective reproduction

Actual condensed v2 console output

```
$ git diff params.yaml
- dense_units: 128
+ dense_units: 256
$ dvc repro
Stage 'prepare' didn't change, skipping
Stage 'preprocess' didn't change, skipping
Running stage 'train': > python src/train.py
Epoch 8/8 - accuracy: 0.8896 - val_accuracy: 0.8932
Running stage 'evaluate': > python src/evaluate.py
{"test_loss": 0.342278, "test_accuracy": 0.8772, "test_samples": 10000}
```



Metrics comparison

Version	Dense units	Test loss	Test accuracy	Change
v1	128	0.348672	87.41%	baseline
v2	256	0.342278	87.72%	+0.31 pp

Only train reran because its dense_units dependency changed. Evaluate reran downstream because the model hash changed. Prepare and preprocess were skipped because their code, parameters, and input hashes were unchanged. The larger hidden layer reduced test loss by 0.006394 and improved accuracy by 0.31 percentage points.

Version tags

```
$ git tag --list -n
v1 Baseline Fashion-MNIST ANN
v2 Fashion-MNIST ANN with 256 dense units
```

Part E simulated two-contributor conflict

Actual merge output

```
$ git merge teammate-sim
Auto-merging dvc.lock
CONFLICT (content): Merge conflict in dvc.lock
Auto-merging metrics.json
CONFLICT (content): Merge conflict in metrics.json
Auto-merging src/preprocess.py
CONFLICT (content): Merge conflict in src/preprocess.py
Automatic merge failed; fix conflicts and then commit the result.
$ git status --short
UU dvc.lock
UU metrics.json
UU src/preprocess.py
```

Source conflict

```
<<<<<< HEAD
scaled = images.astype(float32) / 255.0
return power(scaled, 0.99)
=====
clipped = clip(images.astype(float32), 0, 255)
scaled = clipped / 255.0
return power(scaled, 1.01)
>>>>>> teammate-sim
```

Data conflict

```
dvc.lock contained divergent md5 hashes for:
- data/processed
- models/model.h5
- models/history.csv
- metrics.json

Resolution policy:
1. reconcile source logic
2. choose no stale branch hash
3. regenerate with dvc repro
```

Resolution and reproducibility proof

The merged `normalize` function combines clipping from `teammate-sim` with explicit `float32` division and the original range validation. Instead of choosing either conflicting lockfile, the main-side lock was used only as a parseable starting point and `dvc repro` regenerated the authoritative processed hash. Because the resolved linear transform matched the v2 data exactly, DVC reused the cached train/evaluate outputs.

Final local verification

```
$ dvc repro
Stage 'prepare' didn't change, skipping
Running stage 'preprocess'
Stage 'train' is cached - skipping run, checking out outputs
Stage 'evaluate' is cached - skipping run, checking out outputs
$ dvc status
Data and pipelines are up to date.
$ git log --oneline --graph --all
* ee2871e merge: resolve simulated code and DVC data conflicts
|| * 9a7b3df collab: simulate teammate normalization and data version
* | 47614a4 collab: add independent main normalization and data version
|/
```

Deliverables

Deliverable	Location / proof
GitHub repository	https://github.com/RaoHaris535/Fashion-ANN-Pipeline
Google Drive DVC remote	Folder ID 1iyOk52bAA4H6i9_KrBaXYRkCInOuMYg; 11 files pushed
Final pipeline state	dvc.lock committed; data and pipelines up to date
Submission report	Assignment3_Report_Muhammad_Haris_23L-0807.pdf